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A NOTE ON THE MÖBIUS AND LIOUVILLE FUNCTIONS

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1. Introduction

The purpose of this short note is to investigate the values taken by the Möbius function $\mu(n)$ on consecutive square-free integers, and by the Liouville function $\lambda(n)$ on successive integers. We recall the definition of these two functions:

$\lambda(1)=1$; $\lambda(n)=(-1)^{\Omega(n)}$, where $\Omega(n)$ denotes the number of prime factors of n counted according to multiplicity.

$\mu(n)=\lambda(n)$ if n is square-free (henceforth abbreviated to sf).
$=0$ if $d^2|n$ with $d>1$.

We write q_k for the k-th sf integer. Also we use σ and s to denote $+$ or $-$ signs. With this notation we define

$$(1) \qquad f_{\sigma s}(x)=\sum_{\substack{k\leq x\\ \mu(q_k)=\sigma 1\\ \mu(q_{k+1})=s1}} 1; \quad g_{\sigma s}(x)=\sum_{\substack{n\leq x\\ \lambda(n)=\sigma 1\\ \lambda(n+1)=s1}} 1.$$

Since we have no reason *a priori* to assume the contrary, we may plausibly state the following

CONJECTURE. *For each case* $(s,\sigma)=(+,+),(+,-),(-,-),(-,+)$ *we have*

$$(2) \qquad f_{\sigma s}\sim\frac{x}{4} \quad \textit{and} \quad g_{\sigma s}\sim\frac{x}{4} \quad \textit{as} \quad x\to\infty.$$

We further write $f^*(x)=f_{++}(x)+f_{--}(x)$ and define $g^*(x)$ analogously. Also we put

$$M(X)=\sum_{n\leq X}\mu(n), \quad L(X)=\sum_{n\leq X}\lambda(n).$$

It is then easy to verify the following simple relations between the functions defined in (1):

$$(3) \qquad f_{++}(x)=f_{--}(x)+M(q_x)+\frac{1}{2}(\mu(q_{x+1})-1)$$

$$f_{+-}(x)=f_{-+}(x)+\frac{1}{2}(1-\mu(q_{x+1}))$$

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and similarly for $g_{\sigma s}$ *mutatis mutandis*. It is well-known that

$$M(X) = o(X), \qquad L(X) = o(X), \qquad q_x = \pi^2 x/6 + O(x^{1/2}), \tag{4}$$

(Theorems 333 and 335 of [1] give the first and last of the results in (4) for example). Hence by (3) and (4) it suffices to show that for one pair (σ, s) we have $f_{\sigma s}(x) \sim x/4$ in order to prove our conjecture for all four f functions and similarly for $g_{\sigma s}$. The conjecture (2) is clearly equivalent to the supposition that

$$\sum_{k \leqq x} \mu(q_k)\mu(q_{k+1}) = o(x), \qquad \sum_{n \leqq x} \lambda(n)\lambda(n+1) = o(x).$$

To see this consider, for example, $g_{+-}(x)$ for which we have

$$g_{+-}(x) = \frac{1}{4} \sum_{n \leqq x} (\lambda(n)+1)(1-\lambda(n+1)) = \frac{1}{4}([x] - \sum_{n \leqq x} \lambda(n)\lambda(n+1) + 1 - \lambda([x]+1)).$$

These conjectures are apparently in the same league of difficulty as the prime twins problem so we are very far from proving them! In the rest of the paper, however, we will demonstrate that it is possible to obtain non-trivial bounds for our functions. Our two results are as follows:

THEOREM 1. *For the functions defined by* (1), *and any* $\varepsilon > 0$, *we have*

$$f_{\sigma s}(x) \geqq x(\log x)^{-7-\varepsilon}; \qquad g_{\sigma s}(x) \geqq x(\log x)^{-7-\varepsilon} \tag{5}$$

for $(\sigma, s) = (+, -)$ *and* $(-, +)$ *and all sufficiently large* x.

THEOREM 2. *For the functions defined by* (1) *we have, for* $(\sigma, s) = (+, +)$ *and* $(-, -)$,

$$f_{\sigma s}(x) > \frac{x}{60}(1+o(1)); \quad g_{\sigma s} \geqq \frac{x}{6}(1+o(1)). \tag{6}$$

2. Proof of Theorem 1

First we mention a completely elementary argument. By Theorem 418 of [1] for every $n \geqq 1$ the interval $(n, 2n]$ contains a prime, and so every interval of the form $(2n, 4n]$ for $n \geqq 2$ contains a prime and a square-free integer which is twice a prime. It follows that

$$f_{\sigma s}(x) \gg \log x, \qquad g_{\sigma s}(x) \gg \log x.$$

The somewhat surprising feature of this result is that our proof of Theorem 1 is just as unsophisticated!

In 1979 D. Wolke [6], improving work of D. R. Heath-Brown [3] and Y. Motohashi [4] established, by analytic methods, that almost all intervals of the form

$$[n, n+(\log n)^C)$$

contain sf numbers with two odd prime factors. He gave the value 5.10^6 for C. The "almost all" in the result indicates that the number of integers $\leqq x$ for which the given interval does not contain almost-primes is $o(X)$. More recently, G. Harman

[2] improved the value of C to $7+\eta$ for any $\eta>0$. The important point for our present investigation is that, as opposed to a sieve method [3], the numbers obtained do have precisely two prime factors. It follows that almost all intervals of the form $[n, n+2(\log n)^C)$ contain sf integers of both forms p_1p_2, $2p_3p_4$ (p_j primes). Hence there must be a q_x in the interval with $\mu(q_x)\mu(q_{x+1})=-1$. By the relations (3) the result follows for f_{+-} and f_{-+} taking $\eta=\varepsilon/2$ and x sufficiently large. The result follows *a fortiori* for g_{+-} and g_{-+}. It should be noted that one expects there to be many n in the intervals considered with $\lambda(n)\lambda(n+1)=-1$, and perhaps it is possible to construct an argument that should there only be a few such n in certain intervals then there must be many in other intervals (cf. our proof of Theorem 2 below).

3. Proof of Theorem 2

First we note that analytic methods give results (see [5]) which only imply lower bounds of the form $\sqrt{x}\exp(-3(\log\log x)^{3/2})$ for f_{++}, f_{--} and g_{---}, and no result at all for g_{+++}. However, it was this observation together with the argument of Section 2 which prompted this investigation.

In this section our arguments will again be very elementary. We would like to express our thanks to Dr. R. R. Hall for showing us a simple proof that

$$\sum_{n\leqq X}\lambda(n)\lambda(n+1)\geqq -\frac{2X}{3}+O(1).$$

This was produced as an answer to a question put by E. Bombieri at the Durham Conference on Analytic Number Theory in 1979. Our proof given here was inspired by Hall's proof and leads to the stronger bound

$$\sum_{n\leqq X}\lambda(n)\lambda(n+1)\geqq -\frac{X}{3}+O(\log X). \tag{7}$$

We shall in fact prove that

$$f^*(x)>\frac{x}{30}+O(\sqrt{x}), \tag{8}$$

and

$$g^*(x)\geqq\frac{x}{3}+O(\log x). \tag{9}$$

We note (6) follows from (3), (4), (8) and (9), while (7) follows immediately from (9) (recalling the definition of f^* and g^* in §1).

We start by proving (9). Let us suppose that $\lambda(x)\lambda(x+1)=-1$. Hence $\lambda(2x)\lambda(2x+2)=-1$ so either $\lambda(2x)\lambda(2x+1)=1$ or $\lambda(2x+1)\lambda(2x+2)=1$. By considering all integers $x\in[X, 2X]$ it follows that

$$g^*(4X)-g^*(X)\geqq X+O(1),$$

and so, by induction,

$$g^*(x)\geqq\frac{x}{3}+O(\log x).$$

To use a similar argument for $f^*(x)$ we need to be more careful. The following lemma is a particular example of a well-known result: — *the sf equivalent to the prime k-tuplet conjecture is true.* We give the proof (cf. the proof of Theorem 333 in [1]) for completeness. While writing this paper we learnt that D. R. Heath-Brown was aware of a similar argument.

LEMMA. *Write*

$$P_1(y) = (36y+3)(36y+5)(72y+7),$$

$$P_2(y) = (36y+31)(36y+33)(72y+65).$$

Then the number of sf integers of the form $P_1(y)$ *or* $P_2(y)$ *for* $1 \leqq y \leqq N$ *is*

$$\frac{2N(27-2\pi^2)}{\pi^2}+O(\sqrt{N}).$$

PROOF. We consider integers of the form $P_1(y)$, the proof for $P_2(y)$ follows analogously and, of course, $P_1(y_1)=P_2(y_2)$ has no solutions in integers.

We write

$$A(N) = \sum_{y \leqq N} |\mu(P_1(y))|.$$

Then

$$A(N) = \sum_{1 \leqq y \leqq N} \sum_{d^2 | P_1(y)} \mu(d) = \sum_{d < 9N^{1/2}} \mu(d) \sum_{\substack{y=1 \\ d^2 | P_1(y)}}^{N} 1. \tag{10}$$

If we let $Y(N, d)$ denote the sum over y on the right of (10) we then have $Y(N, 1)=N$; $Y(N, d)=0$ if $3|d$ or $2|d$. On the other hand, by considering congruences modulo d^2, $Y(N, d)=3N/d^2+O(1)$ if $(6, d)=1$ and $d>1$ (note the three factors of $P_1(y)$ are always coprime). Hence

$$A(N) = 3N \sum_{\substack{(d,6)=1 \\ 1 \leqq d \leqq 9N^{1/2}}} \frac{\mu(d)}{d^2} - 2N + O(\sqrt{N}) =$$

$$= 3N \sum_{\substack{d=1 \\ (d,6)=1}}^{\infty} \frac{\mu(d)}{d^2} - 2N + O(\sqrt{N}) =$$

$$= \frac{3N}{\zeta(2)} \frac{4}{3} \frac{9}{8} - 2N + O(\sqrt{N}) =$$

$$= \frac{N(27-2\pi^2)}{\pi^2} + O(\sqrt{N}).$$

This completes the proof of the lemma.

It is now easy to establish (8) and thus complete the proof of Theorem 2. By our lemma the number of sf triplets of the form $36n+3$, $36n+5$, $72n+7$ or $36n+31$, $36n+33$, $72n+65$ less than q_x (recall (4) for the size of q_x) is

$$\frac{(27-2\pi^2)}{216} x + O(\sqrt{x}) > \frac{x}{30} + O(\sqrt{x}).$$

We note that $\mu(m)=0$ for m of the form $36n+4$, $72n+8$, $72n+9$ etc. Now if $\mu(36n+3)\mu(36n+5)=-1$ it follows that either $\mu(72n+6)\mu(72n+7)=1$ or $\mu(72n+7)\mu(72n+10)=1$. A similar argument holds for the triplets of the other form. Thus

$$f^*(x) > \frac{x}{30}+O(\sqrt{x})$$

and the proof is finished. It is, of course, possible to improve the 60 in our result. For example one can consider, in addition to the above argument, sf quintuplets of the form $72y+41$, $72y+42$, $72y+43$, $144y+83$, $144y+85$, since corresponding to each quintuplet there exists a q_x (not of a type given by our previous discussion), with $\mu(q_x)\mu(q_{x+1})=1$.

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